ArrayList<E> :
----

public class ArrayList<E> extends AbstractList<E> implements List<E>, **Cloneable, Serializable**, RandomAccess

* It is an implemented class of List<E> interface available from JDK 1.2V.
* It can accept duplicate, null, homogeneous & hetreogeneous objects.
* It stores the objects based on the index position.
* It uses **"Dynamic Array Data structure"** with default capacity **10**.
* Next capacity of ArrayList would be **16**. [Almost 50% increment]
* Next capacity can be calculated by using the folllowing formula :

New Capacity = (Current Capacity * 3) /2 + 1
= 10 * 3 /2 + 1
= 16

* Methods are not synchronized so, performance wise better than Vector.
* Iterator is Fail Fast Iterator.
* hashCode() & equals() methods are overridden.
* It is mainly used to **retrieve the object** where thread safety is not required.

Constructors in ArrayList :

In ArrayList we have 3 types of Constructors:

1) ArrayList al1 = new ArrayList();
Will create ArrayList object with default capacity 10.

2) ArrayList al2 = new ArrayList(int initialCapacity);
Will create an ArrayList object with user specified Capacity

3) ArrayList al3 = new ArrayList(Collection c)
We can copy any Collection interface implemented class data to the current object reference (Coping one Collection data to another)

//ArrayList Program :

```
package com.ravi.arraylist;

import java.util.ArrayList;
import java.util.List;

//CRUD OPERTAION

record Employee(Integer id, String name, Double salary)
{
}

class Organization
{
   private String name;
   private List<Employee> listOfEmployees;

   public Organization(String name)
   {
    super();
    this.name = name;
    listOfEmployees = new ArrayList<>();  //Composition
   }


   //Create Operation
   public void addEmployee(Employee employee)
   {
       listOfEmployees.add(employee);
       IO.println("Employee Object stored successfully");
   }

   //Read operation
   public void displayListOfEmployee()
   {
       listOfEmployees.forEach(IO::println);
   }


   //Update operation
   public void updateEmployee(Integer id, Double bouns)
   {
       boolean isFound = false;
       for(int i=0; i<listOfEmployees.size(); i++)
       {
           if(listOfEmployees.get(i).id().equals(id))
           {
             listOfEmployees.set(i, new Employee(id, listOfEmployees.get(i).name(),
listOfEmployees.get(i).salary()+bouns));
             IO.println("Employee with id "+id+" is updated!!");
             isFound = true;
             break;
           }
       }

       if(!isFound)
       {
         IO.println("Sorry, Employee with "+id+ " id is not available");
       }
   }


   //delete Operation
   public void deleteEmployee(Integer id)
   {
       listOfEmployees.removeIf(emp -> emp.id().equals(id));
       IO.println("Employee with id "+id+" is removed");
   }
}


public class ArrayListDemo1
{
    public static void main(String[] args)
    {
       Organization wipro = new Organization("Wipro Limited");

       while(true)
       {
           IO.println("Menu for Operation :");
           IO.println("1) Insert Employee Objects :");
           IO.println("2) Display Employee Objects :");
           IO.println("3) Update Employee Object :");
           IO.println("4) Delete Employee Object :");
           IO.println("5) Exit :\n");

           Integer choice = Integer.valueOf(IO.readln("Enter your Choice :"));

           switch(choice)
           {
           case 1->
           {
               Integer id = Integer.valueOf(IO.readln("Enter Employee id :"));
               String name = IO.readln("Emter Employee Name :");
               Double salary = Double.valueOf(IO.readln("Enter Employee Salary :"));
               wipro.addEmployee(new Employee(id, name, salary));

           }

           case 2-> wipro.displayListOfEmployee();


           case 3->
           {
               Integer id = Integer.valueOf(IO.readln("Enter Employee id to update :"));
               Double bonus = Double.valueOf(IO.readln("Enter bonus amount :"));
               wipro.updateEmployee(id, bonus);
           }

           case 4->
           {
               Integer id = Integer.valueOf(IO.readln("Enter Employee id to delete :"));
               wipro.deleteEmployee(id);
           }

           case 5->
           {
               IO.println("Exiting from Application");
               System.exit(0);
           }

           default->
           {
               IO.println("Invalid Choice");
           }

           }

       }

    }
}
```

```
package com.ravi.arraylist;

//addAll() & retainAll()

import java.util.*;
public class ArrayListDemo2
    {
        public static void main(String args[])
        {
          ArrayList<String> al=new ArrayList<>();
          al.add("Virat");
          al.add("Rahul");
          al.add("Rohit");


         Vector<String> v1 = new Vector<>();
         v1.add("Shami");
         v1.add("Bumrah");
         v1.add("Siraj");

         IO.println("Add All");
         al.addAll(v1);

         al.forEach(IO::println);

         IO.println("Retain All");
         ArrayList<String> list = new ArrayList<>();
         list.add("Rohit");
         list.add("Virat");
         list.add("hardik");

         Vector<String> list1 = new Vector<>();
         list1.add("Abhishek");
         list1.add("Head");
         list1.add("hardik");


         list.retainAll(list1);  //common element

         IO.println(list);
   }
}
```

```
package com.ravi.arraylist;

import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;

//Remove duplicate from the List

public class ArrayListDemo3
{
 public static void main(String args[])
 {
     List<Integer> duplicateList = Arrays.asList(1,2,3,4,1,5,3);
     IO.println("Duplicate List :"+duplicateList);

     List<Integer> uniqueList = removeDuplicateElements(duplicateList);
     IO.println("Unique List :"+uniqueList);

 }

 public static List<Integer> removeDuplicateElements(List<Integer> duplicate)
 {
     List<Integer> uniqueList = new ArrayList<>(); //[1 2 3]

     for(Integer num : duplicate)   //num = 1
     {
         if(!uniqueList.contains(num))
         {
             uniqueList.add(num);
         }
     }

     return uniqueList;



 }




}
```

```
package com.ravi.arraylist;

import java.util.ArrayList;
import java.util.List;

/* OLA Ride Application for Customer to print bill based on different Criteria
 * a) For every new Customer first ride is free
   b) Up to 4 KM bill is 80 RS
   c) After 4 KM, for every KM, Customer Need to pay 9 RS Per KM
*/

record Customer(Integer id,String name, String pickupLocation,
         String dropLocation, Integer distance, Long phone)
{

}

class CabService
{
    private String name;
    private List<Customer> listOfCustomers;

    public CabService(String name)
    {
        this.name = name;
        this.listOfCustomers = new ArrayList<>();
    }

    //Helper method
    private boolean isNewCustomer(Customer customer)
    {
        for(Customer cust : listOfCustomers)
        {
            if(cust.id().equals(customer.id()))
            {
                return false;
            }
        }

        return true;
    }

    private void addCustomer(Customer customer)
    {
        listOfCustomers.add(customer);
        IO.println("Customer has registered successfully");
    }

    public double getBill(Customer customer)  //444
    {
        double bill = 0.0;

        if(this.isNewCustomer(customer))
        {
            this.addCustomer(customer);
            IO.println("You are a new customer so, enjoy your first ride...");
            bill = 0.0;
        }
        else if(customer.distance() <=4)
        {
            bill = 80.00;
        }
        else
        {
            bill = 80 + (customer.distance() - 4)* 9;
        }

        return bill;
    }
}

public class ArrayListDemo4
{
    public static void main(String[] args)
    {
        CabService ola = new CabService("OLA");

        Customer c1 = new Customer(111, "Scott", "Ameerpet", "Koti", 12, 9999999999L);
        IO.println("Dear "+c1.name()+ " your bill is :"+ola.getBill(c1));

        c1 = new Customer(111, "Scott", "Ameerpet", "Koti", 12, 9999999999L);
        IO.println("Dear "+c1.name()+ " your bill is :"+ola.getBill(c1));


        c1 = new Customer(111, "Scott", "Ameerpet", "Panjagutta", 3, 9999999999L);
        IO.println("Dear "+c1.name()+ " your bill is :"+ola.getBill(c1));


        Customer c2 = new Customer(222, "Smith", "Ameerprt", "Durgam Cheru", 16,
8888888888L);
        IO.println("Dear "+c2.name()+ " your bill is :"+ola.getBill(c2));

        c2 = new Customer(222, "Smith", "Ameerprt", "Durgam Cheru", 16, 8888888888L);
        IO.println("Dear "+c2.name()+ " your bill is :"+ola.getBill(c2));

        c2 = new Customer(222, "Smith", "Ameerprt", "Moosapet", 4, 8888888888L);
        IO.println("Dear "+c2.name()+ " your bill is :"+ola.getBill(c2));


    }
}
```